

I. INTRODUCTION

ICH Q10 describes one comprehensive model for an effective pharmaceutical quality system that is based on International Organization for Standardization (ISO) quality concepts, includes applicable good manufacturing practice (GMP) regulations, and complements ICH “Q8 Pharmaceutical Development” and ICH “Q9 Quality Risk Management.”³ ICH Q10 is a model for a pharmaceutical quality system that can be implemented throughout the different stages of a product lifecycle. Much of the content of ICH Q10 applicable to manufacturing sites is currently specified by regional GMP requirements. ICH Q10 is not intended to create any new expectations beyond current regulatory requirements. Consequently, the content of ICH Q10 that is additional to current regional GMP requirements is optional.

ICH Q10 demonstrates industry and regulatory authorities’ support of an effective pharmaceutical quality system to enhance the quality and availability of medicines around the world in the interest of public health. Implementation of ICH Q10 throughout the product lifecycle should facilitate innovation and continual improvement and strengthen the link between pharmaceutical development and manufacturing activities. FDA’s guidance documents, including this guidance, do not establish legally enforceable responsibilities. Instead, guidances describe the Agency’s current thinking on a topic and should be viewed only as recommendations, unless specific regulatory or statutory requirements are cited. The use of the word should in Agency guidances means that something is suggested or recommended, but not required.

II. PHARMACEUTICAL QUALITY MANAGEMENT SYSTEM

A. Scope (1.2) This guidance applies to the systems supporting the development and manufacture of pharmaceutical drug substances (i.e., active pharmaceutical ingredients (APIs)) and drug products, including biotechnology and biological products, throughout the product lifecycle. The elements of ICH Q10 should be applied in a manner that is appropriate and proportionate to each of the product lifecycle stages, recognizing the differences among, and the different goals of, each stage (see section IV (3)). For the purposes of this guidance, the product lifecycle includes the following technical activities for new and existing products:

- Pharmaceutical Development
 - o Drug substance development
 - o Formulation development (including container/closure system)
- Manufacture of investigational products
 - o Delivery system development (where relevant)
- Manufacturing process development and scale-up
 - o Analytical method development
- Technology Transfer
 - o New product transfers during development through manufacturing

Transfers within or between manufacturing and testing sites for marketed products •
Commercial Manufacturing o Acquisition and control of materials o Provision of facilities, utilities, and equipment o Production (including packaging and labeling) o Quality control and assurance o Release o Storage o Distribution (excluding wholesaler activities) •
Product Discontinuation o Retention of documentation o Sample retention o Continued product assessment and reporting

B. Relationship of ICH Q10 to Regional GMP Requirements, ISO Standards, and ICH Q7 (1.3) Regional GMP requirements, the ICH guidance “Q7 Good Manufacturing Practice Guidance for Active Pharmaceutical Ingredients,” and ISO quality management system guidelines form the foundation for ICH Q10. To meet the objectives described below, ICH Q10 augments GMPs by describing specific quality system elements and management responsibilities. ICH Q10 provides a harmonized model for a pharmaceutical quality system throughout the lifecycle of a product and is intended to be used together with regional GMP requirements. The regional GMPs do not explicitly address all stages of the product lifecycle (e.g., development). The quality system elements and management responsibilities described in this guidance are intended to encourage the use of science- and risk-based approaches at each lifecycle stage, thereby promoting continual improvement across the entire product lifecycle.

C. Relationship of ICH Q10 to Regulatory Approaches (1.4) Regulatory approaches for a specific product or manufacturing facility should be commensurate with the level of product and process understanding, the results of quality risk management, and the effectiveness of the pharmaceutical quality system. When implemented, the effectiveness of the pharmaceutical quality system can normally be evaluated during a regulatory inspection at the manufacturing site. Potential opportunities to enhance science- and risk-based regulatory approaches are identified in Annex 1. Regulatory processes will be determined by region.

D. ICH Q10 Objectives (1.5) Implementation of the Q10 model should result in achievement of three main objectives that complement or enhance regional GMP requirements. 1. Achieve Product Realization (1.5.1) To establish, implement, and maintain a system that allows the delivery of products with the quality attributes appropriate to meet the needs of patients, health care professionals, regulatory authorities (including compliance with approved regulatory filings) and other internal and external customers. 2 Establish and Maintain a State of Control (1.5.2) To develop and use effective monitoring and control systems for process performance and product quality, thereby providing assurance of continued suitability and capability of processes. Quality risk management can be useful in identifying the monitoring and control systems. 3 Facilitate

Continual Improvement (1.5.3) To identify and implement appropriate product quality improvements, process improvements, variability reduction, innovations, and pharmaceutical quality system enhancements, thereby increasing the ability to fulfill a pharmaceutical manufacturer's own quality needs consistently. Quality risk management can be useful for identifying and prioritizing areas for continual improvement.

E. Enablers: Knowledge Management and Quality Risk Management (1.6) Use of knowledge management and quality risk management will enable a company to implement ICH Q10 effectively and successfully. These enablers will facilitate achievement of the objectives described in section II.D (1.5) above by providing the means for science- and risk based decisions related to product quality. 1. Knowledge Management (1.6.1) Product and process knowledge should be managed from development through the commercial life of the product up to and including product discontinuation. For example, development activities using scientific approaches provide knowledge for product and process understanding. Knowledge management is a systematic approach to acquiring, analyzing, storing, and disseminating information related to products, manufacturing processes, and components. Sources of knowledge include, but are not limited to, prior knowledge (public domain or internally documented); pharmaceutical development studies; technology transfer activities; process validation studies over the product lifecycle; manufacturing experience; innovation; continual improvement; and change management activities. 2. Quality Risk Management (1.6.2) Quality risk management is integral to an effective pharmaceutical quality system. It can provide a proactive approach to identifying, scientifically evaluating, and controlling potential risks to quality. It facilitates continual improvement of process performance and product quality throughout the product lifecycle. ICH Q9 provides principles and examples of tools for quality risk management that can be applied to different aspects of pharmaceutical quality.

F. Design and Content Considerations (1.7) (a) The design, organization, and documentation of the pharmaceutical quality system should be well structured and clear to facilitate common understanding and consistent application. (b) The elements of ICH Q10 should be applied in a manner that is appropriate and proportionate to each of the product lifecycle stages, recognizing the different goals and knowledge available for each stage. (c) The size and complexity of the company's activities should be taken into consideration when developing a new pharmaceutical quality system or modifying an existing one. The design of the pharmaceutical quality system should incorporate appropriate risk management principles. While some aspects of the pharmaceutical quality system can be company wide and others site specific, the effectiveness of the pharmaceutical quality system is normally demonstrated at the site level. (d) The pharmaceutical quality system should include appropriate processes, resources, and

responsibilities to provide assurance of the quality of outsourced activities and purchased materials as described in section III.G (2.7). (e) Management responsibilities, as described in section III (2), should be identified within the pharmaceutical quality system. (f) The pharmaceutical quality system should include the following elements, as described in section IV (3): process performance and product quality monitoring, corrective and preventive action, change management, and management review. (g) Performance indicators, as described in section V (4), should be identified and used to monitor the effectiveness of processes within the pharmaceutical quality system.

G. Quality Manual (1.8) A Quality Manual or equivalent documentation approach should be established and should contain the description of the pharmaceutical quality system. The description should include: (a) The quality policy (see section III (2)). (b) The scope of the pharmaceutical quality system. (c) Identification of the pharmaceutical quality system processes, as well as their sequences, linkages, and interdependencies. Process maps and flow charts can be useful tools to facilitate depicting pharmaceutical quality system processes in a visual manner. (d) Management responsibilities within the pharmaceutical quality system (see section III (2)).

III. MANAGEMENT RESPONSIBILITY

Leadership is essential to establish and maintain a company-wide commitment to quality and for the performance of the pharmaceutical quality system.

A. Management Commitment (2.1) (a) Senior management has the ultimate responsibility to ensure an effective pharmaceutical quality system is in place to achieve the quality objectives, and that roles, responsibilities, and authorities are defined, communicated, and implemented throughout the company. (b) Management should: (1) Participate in the design, implementation, monitoring, and maintenance of an effective pharmaceutical quality system. (2) Demonstrate strong and visible support for the pharmaceutical quality system and ensure its implementation throughout their organization. (3) Ensure a timely and effective communication and escalation process exists to raise quality issues to the appropriate levels of management. (4) Define individual and collective roles, responsibilities, authorities, and inter relationships of all organizational units related to the pharmaceutical quality system. Ensure these interactions are communicated and understood at all levels of the organization. An independent quality unit/structure with authority to fulfill certain pharmaceutical quality system responsibilities is required by regional regulations. (5) Conduct management reviews of process performance and

product quality and of the pharmaceutical quality system. (6) Advocate continual improvement. (7) Commit appropriate resources.

B. Quality Policy (2.2) (a) Senior management should establish a quality policy that describes the overall intentions and direction of the company related to quality. (b) The quality policy should include an expectation to comply with applicable regulatory requirements and should facilitate continual improvement of the pharmaceutical quality system. (c) The quality policy should be communicated to and understood by personnel at all levels in the company. (d) The quality policy should be reviewed periodically for continuing effectiveness.

C. Quality Planning (2.3) (a) Senior management should ensure the quality objectives to implement the quality policy are defined and communicated. (b) Quality objectives should be supported by all relevant levels of the company. (c) Quality objectives should align with the company's strategies and be consistent with the quality policy. (d) Management should provide the appropriate resources and training to achieve the quality objectives. (e) Performance indicators that measure progress against quality objectives should be established, monitored, communicated regularly, and acted upon as appropriate as described in section V.A (4.1) of this document.

D. Resource Management (2.4) (a) Management should determine and provide adequate and appropriate resources (human, financial, materials, facilities, and equipment) to implement and maintain the pharmaceutical quality system and continually improve its effectiveness. (b) Management should ensure that resources are appropriately applied to a specific product, process, or site.

E. Internal Communication (2.5) (a) Management should ensure appropriate communication processes are established and implemented within the organization. (b) Communications processes should ensure the flow of appropriate information between all levels of the company. (c) Communication processes should ensure the appropriate and timely escalation of certain product quality and pharmaceutical quality system issues.

F. Management Review (2.6) (a) Senior management should be responsible for pharmaceutical quality system governance through management review to ensure its continuing suitability and effectiveness. (b) Management should assess the conclusions of periodic reviews of process performance and product quality and of the pharmaceutical quality system, as described in sections IV (3) and V (4).

G. Management of Outsourced Activities and Purchased Materials (2.7) The pharmaceutical quality system, including the management responsibilities described in this section, extends to the control and review of any outsourced activities and quality of

purchased materials. The pharmaceutical company is ultimately responsible to ensure processes are in place to assure the control of outsourced activities and quality of purchased materials. These processes should incorporate quality risk management and include: (a) Assessing prior to outsourcing operations or selecting material suppliers, the suitability and competence of the other party to carry out the activity or provide the material using a defined supply chain (e.g., audits, material evaluations, qualification). (b) Defining the responsibilities and communication processes for quality-related activities of the involved parties. For outsourced activities, this should be included in a written agreement between the contract giver and contract acceptor. (c) Monitoring and review of the performance of the contract acceptor or the quality of the material from the provider, and the identification and implementation of any essential improvements. (d) Monitoring incoming ingredients and materials to ensure they are from approved sources using the agreed supply chain.

H. Management of Change in Product Ownership (2.8) When product ownership changes (e.g., through acquisitions), management should consider the complexity of this and ensure: (a) The ongoing responsibilities are defined for each company involved (b) The essential information is transferred

IV. CONTINUAL IMPROVEMENT OF PROCESS PERFORMANCE AND PRODUCT QUALITY

A. Lifecycle Stage Goals (3.1) The goals of each product lifecycle stage are described below.

1. Pharmaceutical Development (3.1.1) The goal of pharmaceutical development activities is to design a product and its manufacturing process to consistently deliver the intended performance and meet the needs of patients and healthcare professionals, and regulatory authorities and internal customers' requirements. Approaches to pharmaceutical development are described in ICH Q8. The results of exploratory and clinical development studies, while outside the scope of this guidance, are inputs to pharmaceutical development.

2. Technology Transfer (3.1.2) The goal of technology transfer activities is to transfer product and process knowledge between development and manufacturing, and within or between manufacturing sites to achieve product realization. This knowledge forms the basis for the manufacturing process, control strategy, process validation approach, and ongoing continual improvement.

3. Commercial Manufacturing (3.1.3) The goals of manufacturing activities include achieving product realization, establishing and maintaining a state of control, and facilitating continual improvement. The pharmaceutical quality system should assure that the desired product quality is routinely met, suitable process performance is achieved, the

set of controls are appropriate, improvement opportunities are identified and evaluated, and the body of knowledge is continually expanded. 4. Product Discontinuation (3.1.4) The goal of product discontinuation activities is to manage the terminal stage of the product lifecycle effectively. For product discontinuation, a predefined approach should be used to manage activities such as retention of documentation and samples and continued product assessment (e.g., complaint handling and stability) and reporting in accordance with regulatory requirements.

B. Pharmaceutical Quality System Elements (3.2) The elements described below might be required in part under regional GMP regulations. However, the Q10 model's intent is to enhance these elements to promote the lifecycle approach to product quality. These four elements are: • Process performance and product quality monitoring system • Corrective action and preventive action (CAPA) system • Change management system • Management review of process performance and product quality These elements should be applied in a manner that is appropriate and proportionate to each of the product lifecycle stages, recognizing the differences among the stages and the different goals of each stage.

Throughout the product lifecycle, companies are encouraged to evaluate opportunities for innovative approaches to improve product quality. Each element is followed by a table of example applications of the element to the stages of the pharmaceutical lifecycle. 1.

Process Performance and Product Quality Monitoring System 3.2.1 Pharmaceutical companies should plan and execute a system for the monitoring of process performance and product quality to ensure a state of control is maintained. An effective monitoring system provides assurance of the continued capability of processes and controls to produce a product of desired quality and to identify areas for continual improvement. The process performance and product quality monitoring system should: (a) Use quality risk management to establish the control strategy. This can include parameters and attributes related to drug substance and drug product materials and components, facility and equipment operating conditions, in-process controls, finished product specifications, and the associated methods and frequency of monitoring and control. The control strategy should facilitate timely feedback/feedforward and appropriate corrective action and preventive action. (b) Provide the tools for measurement and analysis of parameters and attributes identified in the control strategy (e.g., data management and statistical tools). (c) Analyze parameters and attributes identified in the control strategy to verify continued operation within a state of control. (d) Identify sources of variation affecting process performance and product quality for potential continual improvement activities to reduce or control variation. (e) Include feedback on product quality from both internal and external sources (e.g., complaints, product rejections, nonconformances, recalls, deviations, audits and regulatory inspections, and findings). (f) Provide knowledge to enhance process

understanding, enrich the design space (where established), and enable innovative approaches to process validation.

Table I: Application of Process Performance and Product Quality Monitoring System Throughout the Product Lifecycle

Pharmaceutical Development	Technology Transfer	Commercial Manufacturing	Product Discontinuation
Process and product knowledge generated and process and product monitoring conducted throughout development can be used to establish a control strategy for manufacturing.	Monitoring during scale-up activities can provide a preliminary indication of process performance and the successful integration into manufacturing. Knowledge obtained during transfer and scale-up activities can be useful in further developing the control strategy.	A well-defined system for process performance and product quality monitoring should be applied to assure performance within a state of control and to identify improvement areas.	Once manufacturing ceases, monitoring such as stability testing should continue to completion of the studies. Appropriate action on marketed product should continue to be executed according to regional regulations.

Table II: Application of Corrective Action and Preventive Action System Throughout the Product Lifecycle

Pharmaceutical Development	Technology Transfer	Commercial Manufacturing	Product Discontinuation
Product or process variability is explored. CAPA methodology is useful where corrective actions and preventive actions are incorporated into the iterative design and development process.	CAPA can be used as an effective system for feedback, feedforward, and continual improvement.	CAPA should be used, and the effectiveness of the actions should be evaluated.	CAPA should continue after the product is discontinued. The impact on product remaining on the market should be considered, as well as other products that might be affected.

Change Management System : Innovation, continual improvement, the outputs of process performance and product quality monitoring, and CAPA drive change. To evaluate, approve, and implement these changes properly, a company should have an effective change management system. There is generally a difference in formality of change management processes prior to the initial regulatory submission and after submission, where changes to the regulatory filing might be required under regional requirements. The change management system ensures continual improvement is undertaken in a timely and effective manner. It should provide a high degree of assurance there are no unintended consequences of the change. The change management system should include the

following, as appropriate for the stage of the lifecycle: (a) Quality risk management should be utilized to evaluate proposed changes. The level of effort and formality of the evaluation should be commensurate with the level of risk. (b) Proposed changes should be evaluated relative to the marketing authorization, including design space, where established, and/or current product and process understanding. There should be an assessment to determine whether a change to the regulatory filing is required under regional requirements. As stated in ICH Q8, working within the design space is not considered a change (from a regulatory filing perspective). However, from a pharmaceutical quality system standpoint, all changes should be evaluated by a company's change management system. (c) Proposed changes should be evaluated by expert teams contributing the appropriate expertise and knowledge from relevant areas (e.g., Pharmaceutical Development, Manufacturing, Quality, Regulatory Affairs, and Medical) to ensure the change is technically justified. Prospective evaluation criteria for a proposed change should be set. (d) After implementation, an evaluation of the change should be undertaken to confirm the change objectives were achieved and that there was no deleterious impact on product quality.

Table III: Application of Change Management System Throughout the Product Lifecycle

Pharmaceutical Development	Technology Transfer	Commercial Manufacturing	Product Discontinuation
Change is an inherent part of the development process and should be documented; the formality of the change management process should be consistent with the stage of pharmaceutical development.	The change management system should provide management and documentation of adjustments made to the process during technology transfer activities.	A formal change management system should be in place for commercial manufacturing. Oversight by the quality unit should provide assurance of appropriate science- and risk-based assessments.	Any changes after product discontinuation should go through an appropriate change management system.

4. Management Review of Process Performance and Product Quality (3.2.4) Management review should provide assurance that process performance and product quality are managed over the lifecycle. Depending on the size and complexity of the company, management review can be a series of reviews at various levels of management and should include a timely and effective communication and escalation process to raise appropriate quality issues to senior levels of management for review. (a) The management review system should include: (1) The results of regulatory inspections and findings, audits and other assessments, and commitments made to regulatory authorities (2) Periodic quality reviews, that can include: (i) Measures of customer satisfaction such as product quality complaints and recalls (ii) (iii) Conclusions of process performance

and product quality monitoring The effectiveness of process and product changes including those arising from corrective action and preventive actions (3) Any follow-up actions from previous management reviews (b) The management review system should identify appropriate actions, such as: (1) Improvements to manufacturing processes and products (2) Provision, training, and/or realignment of resources (3) Capture and dissemination of knowledge

Table IV: Application of Management Review of Process Performance and Product Quality Throughout the Product Lifecycle

Pharmaceutical Development	Technology Transfer	Commercial Manufacturing	Product Discontinuation
Aspects of management review can be performed to ensure adequacy of the product and process design.	Aspects of management review should be performed to ensure the developed product and process can be manufactured at commercial scale.	Management review should be a structured system, as described above, and should support continual improvement.	Management review should include such items as product stability and product quality complaints.